

Project Design Phase

Solution Architecture

Date	28 September 2026
Team ID	SWTID-2026-2588
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

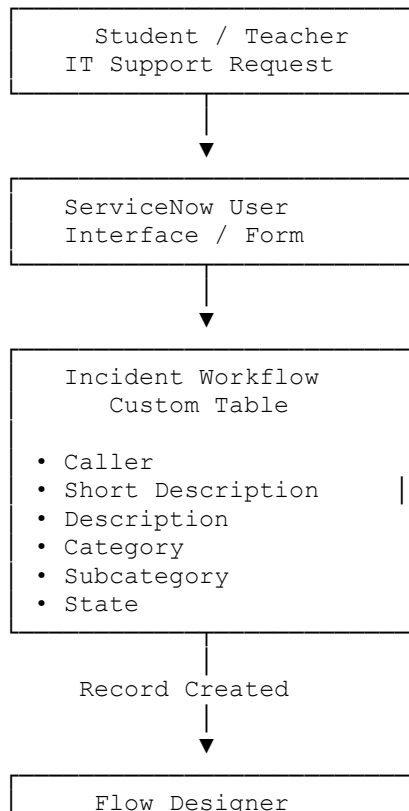
Solution Architecture:

The proposed **Auto Ticket Classification using Flow Designer** system is implemented on the ServiceNow platform. The architecture consists of a user interface for submitting IT support requests, a custom **Incident Workflow** table for storing ticket information, and **Flow Designer** for automating ticket classification and notification.

When a student or teacher creates an IT ticket, the request is stored in the Incident Workflow table. The Flow Designer flow is triggered when a new record is created and the Category field is empty. The flow evaluates predefined keywords in the ticket description and automatically determines the appropriate Category and Subcategory.

After classification, the ticket record is updated with the identified values. An automated email is then sent to the caller confirming that the request has been submitted. The classified ticket becomes available to the IT helpdesk for further processing.

2. Solution Architecture Diagram



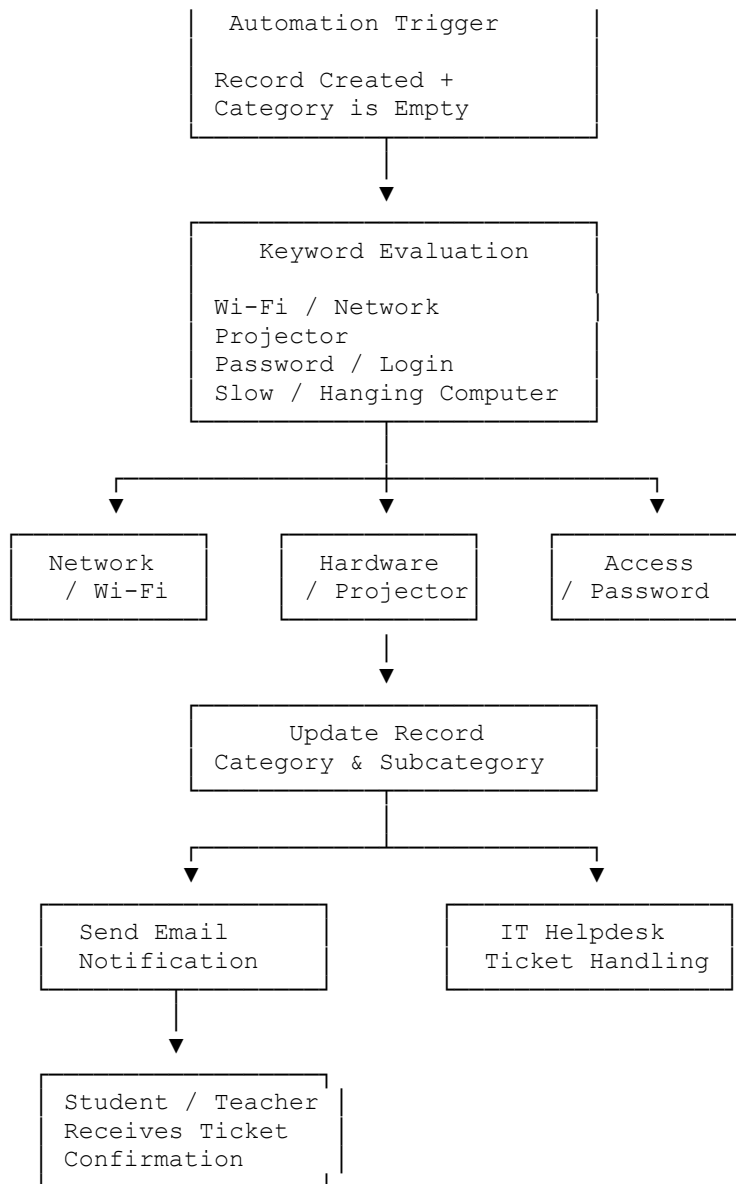


Figure 1: Solution Architecture and Data Flow of Auto Ticket Classification using Flow Designer

3. Architecture Components

Component	Technology / Feature	Purpose
User Interface	ServiceNow Forms	Allows students/teachers or authorized users to create IT tickets
Ticket Management	Incident Workflow Custom Table	Stores ticket details in a structured format
Automation Engine	Flow Designer	Automates classification and notification
Trigger	Record Created	Starts the flow whenever a new ticket is created
Classification Logic	If / Else If Conditions	Checks predefined keywords in the issue description
Category Management	Choice Fields	Stores Network, Hardware, Access, and Performance

Subcategory Management	Choice Fields + Dependency	Maintains valid category–subcategory combinations
Record Update	Update Record Action	Writes the classified values into the ticket
Notification	Send Email Action	Sends confirmation to the caller
Platform	ServiceNow Cloud	Provides the application, database, workflow, and security infrastructure

4. Classification Flow

The classification component uses predefined rules rather than a machine-learning model.

Detected Keyword / Pattern	Category	Subcategory
Wi-Fi, Network	Network	Wi-Fi
Projector	Hardware	Projector
Password, Login, Forgot Password	Access	Forgot Password
Slow, Hanging, Slow Computer	Performance	Slow Computer

5. Data Flow

The overall data flow of the solution is:

Student/Teacher → IT Ticket Form → Incident Workflow Table → Flow Designer → Keyword Evaluation → Category & Subcategory Update → Email Notification → IT Helpdesk

The architecture separates ticket data storage from workflow processing, making the system easier to maintain and extend. Additional classification rules, notifications, assignment logic, and SLA-related workflows can be added later without changing the basic ticket submission process.

6. Architecture Benefits

- **Automation:** Reduces manual ticket classification.
- **Consistency:** Applies the same predefined classification rules to similar requests.
- **Efficiency:** Allows IT staff to focus on resolving issues rather than categorizing tickets.
- **Data Accuracy:** Uses predefined categories and dependent subcategories.
- **User Communication:** Provides automatic confirmation through email.
- **Maintainability:** Flow Designer allows workflow rules to be modified without extensive scripting.
- **Scalability:** Additional issue types and automation rules can be added as the school's IT requirements grow.